



A novel patentability detection model based on Siamese network

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Abstract

Patentability assessment is the most critical stage of the patent registration process. For an invention to be granted a patent, it must meet the patentability criteria of novelty, inventive step, and industrial applicability. Evaluating the novelty and inventive step requires a comparison of patent texts in terms of their technical scope. This comparison is conducted by experts from authorized patent offices and communicated to applicants through official reports. Under the EP and PCT conventions, patentability evaluations are expressed in search reports using the categories X, Y, and A. Due to the lengthy and complex structure of patent texts, as well as the specialized nature of novelty and inventive step comparisons, patentability assessment is a challenging and time-consuming process. Consequently, patent registration timelines are extended, and inventions are often submitted prematurely, before reaching a patentable level of development. The aim of this study is to address the patentability assessment problem from a unique perspective, distinguishing it from other semantic similarity tasks, by proposing a semantic similarity model specifically focused on patentability evaluation. The proposed model generates patentability text pairs with a focus on patent claims and determines the similarity between these pairs using the X, Y, and A categorization. The experiment of the proposed model shows that the integration of the Siamese network structure with CNN and LSTM deep learning architectures has proven to play a highly effective role in achieving accurate results. By utilizing the X, Y, and A codes identified by experts in published patent search reports for model training, this study marks a significant step forward in solving the patentability assessment problem.

Keywords Patent · Patentability · Semantic similarity · Deep learning · Siamese network

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Introduction

With the increasing strategic importance of patent rights in the technology ecosystem, the number of patent applications is steadily rising (Bonino et al., 2010), and the volume of technical knowledge contained in patent databases has reached immense proportions (Schwander, 2000; Björklund, 1991). Patent documents contain unique, up-to-date information that is often unavailable in other sources (Allen & Oppenheim, 1979; Asche, 2017). R&D activities that do not leverage the insights from patent literature frequently fall short of the desired quality (Verhulst, 1997; Atal & Bar, 2010), and many scientific publications have used patents as evidence to demonstrate the state of the art (Li et al., 2013; Chiu & Su, 2014; Yuan & Kumiko, 2014). In addition to serving as a repository of technical advancements, patent resources are becoming increasingly attractive for data analysis and knowledge extraction (Lin & Chou, 2025). As a large-scale, real-world dataset, patent information holds significant potential for conducting research and experiments aimed at solving a wide range of problems. It can also play a crucial role in addressing key challenges in the patenting process. Among these, the most critical aspect to streamline is patentability assessment, the evaluation of whether an invention meets the novelty and inventive step criteria by comparing it with the most relevant prior art.

In practice, patentability assessment is performed by examiners and they report their findings through X, Y, and A codes in search reports under both the European Patent Convention (EPC) and the Patent Cooperation Treaty (PCT) frameworks (Loveniers, 2018). The process of conducting patentability assessments and preparing search reports is both time-consuming and requires significant expertise, often resulting in delays in the patent registration process (Frakes & Wasserman, 2015). Many inventions are submitted for patent applications without meeting patentability criteria, as inventors are unable to conduct patentability assessments themselves. Consequently, these applications fail to secure patent protection (Gaudry, 2012). The development of models capable of performing patentability assessments using emerging data science and deep learning techniques could play a pivotal role in addressing this issue. To design a model capable of effectively handling patentability assessment, it is essential to first develop a deep understanding of the unique characteristics and intrinsic dynamics of the patentability evaluation process itself.

Semantic similarity techniques, which are widely employed to identify the most relevant prior art, form the foundation for developing automated approaches to patentability assessment (Shalaby & Zadrozny, 2019). However, the criteria of novelty and inventive step introduce domain-specific requirements that go beyond generic similarity measures (Schellekens, 2022). It should be clearly emphasized that patentability assessment is fundamentally distinct from other forms of semantic comparisons, as it involves not only identifying prior art but also evaluating whether an invention satisfies the novelty and inventive step criteria under patent law (Clarke, 2018). That is, identifying a document that is semantically similar to a patent application is not the same as determining whether the application satisfies the criteria for patentability. Moreover, patentability evaluation is entirely focused on patent claims, as they represent the core of the assessment process. Solely conducting semantic similarity comparisons between patent descriptions will not yield accurate results for determining patentability. These distinctions illustrate that patentability assessment goes far beyond simple semantic similarity; it requires a nuanced, domain-specific evaluation of technical content in light of legal criteria.

Although significant progress has been made in studies focusing on the semantic comparison of patents, research specifically aimed at patentability assessment remains

relatively scarce (Kayakökü & Tüfekci, 2025). Existing approaches rarely address inventive step evaluation due to the complexity of capturing nuanced technical contributions beyond mere textual similarity. Most studies in the literature focus primarily on identifying the most similar patents, while those addressing patentability tend to concentrate solely on novelty assessments. Moreover, semantic comparisons of patents are often conducted with a focus on the meaning of the entire text, whereas the claims, which constitute the core of patentability assessment, remain outside the primary focus of such processes. This misalignment limits the applicability of existing semantic models for actual patent examination practices and has hindered the development of fully automated tools for comprehensive patentability assessment. There is an urgent need for semantic similarity models capable of evaluating patentability with attention to its domain-specific characteristics, encompassing both novelty and inventive step, which remains a critical and underexplored gap in the field.

The main contribution of this study is the development of a deep learning–based framework that not only measures semantic similarity but also performs novelty and inventive step assessments using real-world patent search report data. The proposed deep learning architecture is trained on published search reports to identify semantic relationships between patent claims and prior art, categorized according to the X, Y, and A classification scheme. In this paper first, related work in the field of patent text analysis is reviewed, followed by a detailed outline of the proposed model and its patentability assessment capabilities. Subsequently, test results conducted using a patentability dataset, specifically developed for the testing of the model in the field of electric vehicles, are presented.

Related works

Before reviewing the existing studies on patentability assessment, it is essential to clarify its position within the broader spectrum of research conducted on patent data and to accurately define its relationship with other tasks in this domain. Patent data contains both structured information such as filing dates, patent classifications, and citation records, and unstructured textual content, including descriptions, claims, and abstracts (Abbas et al., 2014; Madani & Weber, 2016). Analyses conducted on this multifaceted data often draw on a range of techniques, including text mining, natural language processing (NLP), artificial intelligence (AI), and deep learning (Arts et al., 2018; Chandrasekaran & Mago, 2021).

Beyond patentability assessment, research on patent data encompasses a variety of objectives, such as patent classification, clustering, citation analysis, patent network mapping, bibliometric evaluation, patent valuation, cross-lingual patent analysis, document summarization, technological trend detection, technology forecasting, R&D planning, strategic planning, competitor profiling, and partner selection (Bonino et al., 2010; Chen et al., 2015; Chiavetta & Porter, 2013; Rodriguez-Esteban & Bundschus, 2016). While these tasks share methodological overlap, particularly in the use of semantic comparison techniques, they differ significantly in their objectives, required domain expertise, and evaluation frameworks (Kayakökü & Akay, 2021; Walter et al., 2017).

Patentability assessment represents the most critical task within the broader category of processes that involve examining the semantic similarity of patent texts. Semantic similarity methods are widely applied in tasks such as prior art retrieval, patent clustering, technology forecasting and so on. Beyond these applications, patentability assessment also benefits from semantic comparison techniques; however, it requires the integration of

additional domain-specific and legal considerations related to novelty and inventive step criteria. As explained in the introduction, patentability assessment should be regarded as a highly specialized and legally contextualized form of semantic comparison applied to patent texts.

In the literature, only a limited number of semantic similarity studies have explicitly focused on patentability evaluation. Nevertheless, there is a considerable body of work in which semantic comparisons between patents have been conducted for other purposes. Awareness of these studies is valuable, as methods originally developed for different objectives can inspire or be adapted to semantic comparisons designed specifically for patentability assessment. For this reason, this section first reviews studies directly focused on patentability assessment, and then, in a separate subsection, examines works that investigate semantic similarity of patents for other purposes but whose methodologies may be transferable to patentability evaluation.

Patentability assessment oriented studies

Patentability assessment refers to the evaluation of whether an invention described in a patent application meets the legal requirements of novelty and inventive step when compared to the prior art. This process is distinct from other forms of semantic comparison, as it goes beyond identifying textual similarity to incorporate legal reasoning and domain-specific interpretation under patent law. Patent examiners conduct this evaluation during the examination stage, and only inventions satisfying both criteria are granted patents (Clarke, 2018). Importantly, the assessment is centered exclusively on the claims, which define the legal scope of protection; semantic similarity between descriptions alone is insufficient for determining patentability. The text subject to evaluation is therefore the precise content of each claim (Lahorte, 2018).

In practice, examiners document their reasoning through formal search reports, which classify cited prior art under the European Patent Convention (EPC) and the Patent Cooperation Treaty (PCT) frameworks. These reports apply standardized codes: X for documents that independently anticipate a claim (indicating lack of novelty or inventive step), Y for those that preserve novelty but render the claim obvious in combination with other Y documents (indicating lack of inventive step), and A for documents that are technically relevant but do not affect patentability (Loveniers, 2018). This coding system is a valuable but under-utilized resource for developing automated models that emulate examiner decision-making.

Despite the centrality of patentability assessment in the patent system, the number of studies directly addressing it is relatively small. Many works on patent text analysis avoid explicit novelty and inventive step evaluation, likely due to the legal complexity and the domain-specific expertise required. For instance, a recent meta-analysis found that only about 20% of studies on semantic comparison of patents explicitly reference the concept of patentability (Kayakökü & Tüfekci, 2025).

As a widely observed trend in the literature, some studies have focused only on the novelty criterion of patentability. For example, Jeon et al. (2022) vectorized patent documents using the Doc2Vec method and applied the Local Outlier Factor algorithm to assess novelty, without addressing the inventive step criterion. Similarly, Chikkamath et al. (2020) conducted an empirical study integrating machine learning and natural language processing techniques, comparing the performance of various word embedding models solely for novelty detection, again neglecting the inventive step. In another study, Schmitt et al.

(2024) attempted to model the novelty criterion using deep learning algorithms. However, they concluded that the assessment of inventive step requires a more detailed and sophisticated approach.

One of the few studies that directly focus on patentability assessment is the work by Schmitt et al. (2023), who proposed a mathematical-logical framework for evaluating patentability criteria by representing patent claims as structured combinations of technical features and searching for matching structures in the prior art. While their approach provides a formal model of patentability reasoning, it does not utilize expert-labeled training data or pursue automated decision-making through deep learning. Similarly, Lo and Chu (2021) employed USPTO office actions to train a multi-label classifier that maps patent claims to statutory rejection grounds. However, their model does not involve semantic comparison between claims and cited texts, which is central to novelty and inventive step evaluation.

Miyaoka and Akakura (2019) focused on predicting grant or rejection outcomes using statistical attributes of patent descriptions through SVM models, without modeling the comparative reasoning between examined and cited documents in term of both novelty and inventive step. Zhang et al. (2017), on the other hand, developed a retrieval and summarization framework to assist experts in identifying and reviewing relevant prior art but did not aim to automatically infer patentability. Schellekens (2022) focused on the inventive step criterion and pointed out that evaluating the inventive step is a multidimensional and nuanced process, and even AI techniques fall short in demonstrating an inventive approach.

In contrast to these studies, the model proposed in this paper is trained directly on expert-assigned X, Y, and A labels extracted from formal patent search reports. It constructs semantically aligned text pairs to emulate how examiners compare claim content with relevant prior art and uses Siamese neural networks to learn these decision patterns, offering a novel and scalable approach to automating patentability assessment.

Semantic comparisons of patent texts for other purposes

Although the following studies were not designed directly for patentability assessment, their methodologies for detecting semantic similarity between patent documents can be adapted or extended to support novelty and inventive step evaluation.

Text mining and integration of artificial intelligence

For analyzing semantic similarity of patent texts, comparison methods based on set theory, where patent documents are defined as clusters of technical concepts (Moehrle, 2010; Park et al., 2012), are widely used. Similarly, vector space models, which define patent documents as vectors based on the technical concepts they contain, are also prevalent (Shahmirzadi et al., 2019; Wang & Chen, 2019).

Efforts to improve semantic comparisons have included studies that use the subject-action-object (SAO) approach to identify technical concepts in patent documents, resulting in more accurate semantic comparisons (An et al., 2021; Wang et al., 2019; Bergmann et al., 2008). Furthermore, the application of topic modeling techniques such as Latent Semantic Analysis (LSA) and Latent Dirichlet Allocation (LDA) has enhanced the performance of semantic comparison methods (Wang & Chen, 2019; Moehrle, 2019; Cvitanić et al., 2016).

In addition to corpus-based methods, knowledge-based approaches, which leverage pre-existing resources such as ontologies, are also employed for the semantic comparison of patents (Indukuri et al., 2007; Sharma et al., 2015, 2016). Some researchers have gone further by generating new semantic ontologies specifically tailored to the patent domain using patent texts as their source (Jang et al., 2021; Sarica et al., 2020).

Although these methods have yielded significant results in identifying the semantic similarities of patent texts, they often fall short when dealing with the complex structure and technical terminology of patent documents. However, combining these methods with artificial intelligence (AI) techniques leads to more effective outcomes (Genin & Zolkin, 2021; Villa & Wirz, 2022).

Setchi et al. (2021) emphasized the challenges of patent research and underlined the importance of AI techniques in overcoming these difficulties. Integrating AI into patent research processes has been particularly successful in determining the patent class of an invention and accelerating the process of accessing relevant patent documents.

Deep learning methods

At this point, deep learning algorithms and methods also offer the potential to overcome these challenges by delivering surprisingly effective results in data analysis. Through the application of deep learning methods in patent analysis, researchers have begun to achieve satisfactory results from various perspectives (Aristodemou & Tietze, 2018; Krestel et al., 2021).

Whalen et al. (2020) noted that both the Word2Vec method and its extension, Doc2Vec, are widely used in determining the semantic similarity of patents. A study by Helmers et al. (2019) analyzed patent sections such as the description, abstract, and claims separately to compare the performance of methods like TF-IDF, Latent Semantic Analysis (LSA), Word2Vec combined with Bag of Words (BoW) and Doc2Vec, which was more effective in analyzing abstracts and claims.

Aras et al. (2018) demonstrated that a Word2Vec model trained specifically on patent documents significantly outperforms a general Word2Vec model in measuring patent similarity, as it better captures the unique technical terminology and complex structure of patent texts. Krishna et al. (2019) utilized deep learning algorithms, including FastText and Word2Vec, to identify synonyms and semantically related keywords in patent documents, finding that FastText outperformed Word2Vec and resulting in the generation of a dataset of synonymous and related terms validated by patent experts.

In another study, Kim et al., (2020a, 2020b) proposed the Sao2Vec method, which vectorizes subject-action-object (SAO) features of patent documents using Doc2Vec, achieving 3% better performance than the classic Doc2Vec and 115% improvement over frequency-based SAO methods.

Lu et al. (2020) demonstrated that combining Word2Vec with LSTM for context embedding and processing the vectors through CNN and MLP significantly improved patent similarity detection, achieving 94% accuracy compared to 69% with traditional TF-IDF+Cosine Similarity methods. Kim et al., (2020a, 2020b) demonstrated the efficiency of deep learning in patent mining by achieving 97.61% accuracy using a Doc2Vec+DEC (Deep Embedding Clustering) method, outperforming other approaches such as Bag of Words+K-means (55.47%) and Doc2Vec+GMM (97.55%).

Lei et al. (2019) highlighted the limitations of traditional vector space models, such as Bag of Words and TF-IDF, in capturing semantic relationships and sentence

structures in patent documents, and proposed feature extraction using CNNs to generate more accurate and meaningful vector representations, achieving a 91% accuracy compared to 82% with traditional methods. Chung and Sohn (2020) applied a hybrid approach for the early identification of valuable patents by classifying the abstract and claims sections of patent documents using a combined CNN and bi-LSTM model, effectively addressing the limitations of using CNN or RNN alone and achieving high performance.

In another study, 1010 patent abstracts were analyzed with a hybrid approach, combining BiLSTM + CRF for entity recognition and BiGRU + HAN for semantic relationship extraction. By leveraging RNN variants (LSTM and GRU) with CRF for grammar tag prediction and HAN for document classification, the proposed method outperformed previous approaches, achieving 45.8% precision, 58.8% recall, and a 51.5% F-score (Chen et al., 2020). Choi et al. (2022) proposed a two-stage deep learning framework combining CSNet for candidate selection and CRNet for relevance ranking, achieving improved prior art search performance with an MRR score of 0.2506.

Transformer-based and advanced neural approaches

Recent studies highlight the use of Transformer neural networks, such as GPT, GPT-2, and BERT, in analyzing patent texts, as these architectures outperform other deep learning models in understanding and processing textual data, making them highly effective for patent analysis (Lee, 2019).

Lee and Hsiang (2020) combined GPT-2 and BERT models for patent similarity detection, achieving promising results but highlighting challenges with long patent texts. Similarly, Stamatis (2022) achieved promising results in prior art searches by employing a two-stage approach combining BM25 and BERT models.

In another study, Li et al. (2023) focused on patent text classification, addressing feature extraction challenges by utilizing a dictionary network combined with Word-2Vec, CNN, LSTM, and BERT models, with BERT achieving the best results.

In a study by Nemani and Vollala (2022), leveraging the capabilities of Transformer deep learning networks, such as self-attention and non-sequential data processing, DeBERTa variants were employed to detect semantic similarity in patent texts, with the DeBERTa-Small model achieving the highest performance.

Despite these advances, most AI-based semantic comparison studies in the patent domain remain task-agnostic, meaning they are not explicitly optimized for patentability assessment. While they can effectively retrieve similar documents or classify patents, they typically lack the integration of legal reasoning needed to determine novelty and inventive step. As such, their direct applicability to patentability evaluation is limited without incorporating examiner-driven labeling frameworks such as the X, Y, and A codes described in Sect. “[Patentability assessment oriented studies](#)”.

The approach proposed in this study bridges this gap by combining Siamese neural networks with a novel, real-world examiner-labeled training dataset specifically developed for this purpose, thereby enabling semantic similarity analysis that is legally contextualized for patentability assessment. This integration not only leverages the strengths of deep learning in handling complex patent language but also ensures that the output aligns with established legal standards in patent examination.

Proposed model

The proposed model evaluates the patentability of a given patent document by comparing it with other patents within the same technical field. For training the deep learning architecture, the model leverages published patent search reports from the relevant technical domain. These reports serve as a critical resource for determining how the compared patent documents influence the patentability of claims, categorized into X, Y, and A classes. The overall structure of the model is depicted in Fig. 1, and the subsequent sections will provide a comprehensive explanation of the model's design phases and implementation.

Generation of the patentability dataset

The model's key feature is its use of domain-specific datasets for patentability analysis, generated from published patent search reports containing X, Y, and A labels. A patent search report containing the labels X, Y, and A is shown in Fig. 2. These labels reflect patent examiners' assessments, enabling the generation of specialized training data tailored to the technical field of the patent under evaluation. This approach aligns with the principle that patent examiners can only assess inventions within their expertise, particularly for the inventive step criterion, emphasizing the model's reliance on domain-specific expertise to ensure accuracy in patentability decisions.

As seen on Fig. 1 in order to construct the dataset from the search reports, relevant documents must be retrieved from patent search engines using keywords and/or IPC/CPC classifications aligned with the technical domain of the patent under review. Specifically, reports containing X, Y, and A codes—critical for training of deep learning architecture—are obtained by querying EP (European Patent) or WO (World Intellectual Property Organization) publications marked with A1 or A3 codes, as these reports include the necessary labels. Searches employing “keywords and/or IPC classification” combined with “EP or WO” and “A1 or A3” yield the desired results. Extracting essential information such as publication numbers, evaluated claim numbers, cited patents, and similarity categories (X, Y, A) from these typically PDF reports is achieved using OCR or similar text-recognition technologies.

Once this information is extracted, the patent description of the investigated patent can be accessed using its publication number, the claim text can be retrieved using the claim number, and the description of the cited patent can be obtained using the publication number of the cited patent. By combining this information with the similarity category (X, Y,

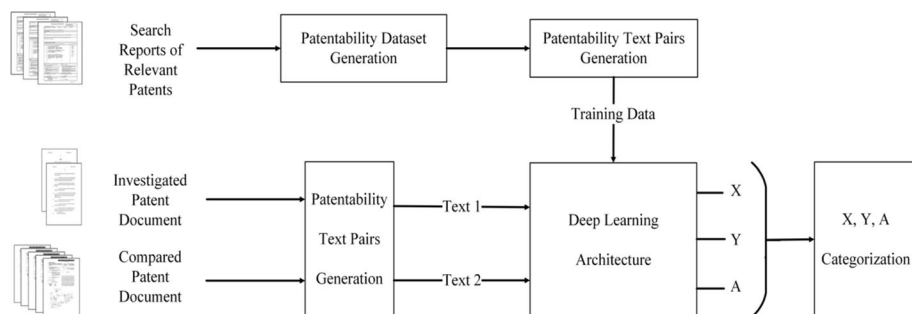


Fig. 1 The design of the proposed model

EP 3 805 039 A1



EUROPEAN SEARCH REPORT

Application Number
EP 20 17 9729

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X	US 2019/014718 A1 (UEMURA KATSUHIKO [JP] ET AL) 17 January 2019 (2019-01-17)	1-5	INV. B60L53/80
Y	* figures 1,3-13 * * paragraphs [0107], [0117] - [0120] *	2,3,5	ADD. H01R13/629 H01R13/72 B60K1/04
X	CA 2 242 337 A1 (RESERVATET I HOVAS HB [SE]) 7 August 1997 (1997-08-07)	1,4	
Y	* figures 1-4 * * page 1, lines 2-7 *	2,3,5	
A	DE 10 2004 047339 A1 (LINDE AG [DE]) 30 March 2006 (2006-03-30)	1-5	
A	WO 2012/176350 A1 (YANMAR CO LTD [JP]; MORIGUCHI KAZUMA [JP]; EBIHARA TOMOYUKI [JP]) 27 December 2012 (2012-12-27)	1-5	

Fig. 2 An example of patent search report

or A), a patentability dataset can be generated. This dataset will consist of four key columns; the description of the investigated patent, the text of the claim being evaluated for patentability, the description of the cited patent and the patentability category (X, Y, or A) between evaluated claim of the investigated patent and cited patent. These four columns reflect the patent examiner's patentability evaluation, providing a robust dataset for training the model.

Generation of patentability text pairs

In the design of the model, evaluating entire patent descriptions for patentability would demand high computational resources and capacity. To optimize the model's performance and efficiency, it is more effective to extract and compare only the most relevant sections of these descriptions. This approach aligns with how patent examiners focus on the parts of the description that are most closely related to the claims in question, as it is impractical for them to read the entire description, especially when it includes details unrelated to the claims being assessed for patentability.

The model filters and extracts text pairs based on claimed elements, ensuring relevance to patentability. Claims and description paragraphs are vectorized, and the most similar paragraphs to the claims are selected from both the examined and cited patents. By comparing either the claim to the extracted section of the cited description or sections from

the two descriptions, the model will focus its patentability assessment on these text pairs. These pairs form the foundation for the subsequent data preparation and model training steps illustrated in Fig. 1.

For clarity, throughout this study, the terminology of “claim,” “description,” and “cited description” is used consistently to avoid ambiguity. The term claim always refers to the first independent claim of the current investigated patent application. The term description (without any modifier) refers to the description of the same investigated patent. When the phrase cited description, it refers to prior art documents cited in patent search reports. Importantly, in this study there is no mentioning of the claims of cited patents; only their descriptions are considered. This naming convention establishes a clear distinction between the investigated patent’s internal components and the prior art used for comparison. Accordingly, in the following sections the constructed patentability text pairs may alternatively be called as claim-to-cited description or description-to-cited description pairs. This consistent naming convention ensures that the dataset design remains transparent and unambiguous for interpreting model performance.

Data preparation for deep learning architecture

Once the patentability text pairs are generated, the next step is to prepare them for the deep learning architecture. A deliberate choice was made to process patent texts in their original form rather than applying heavy preprocessing, such as removing stopwords, filtering reference numbers, or extracting key terms. This is because in patent claims every word can influence the meaning and scope, and empirical testing confirmed that preserving full text yields higher accuracy.

The text pairs are transformed into vector representations using advanced embedding methods. While traditional approaches like Bag of Words, TF-IDF, LSA, and LDA are available, modern neural embeddings such as Word2Vec, Doc2Vec, and FastText offer more context-awareness. However, transformer-based models, particularly BERT, have shown superior performance in capturing semantic nuances in patent language. In this study, the BERT-Base-uncased model is employed for an optimal trade-off between accuracy and computational efficiency, with additional experiments conducted using BERT for Patents to assess improvements in domain-specific understanding.

The resulting patentability text pairs, derived from the curated patent dataset, are then used as the primary training data for the deep learning architecture. Through exposure to these labeled pairs, the model learns to determine the degree of semantic similarity between compared patent documents within the real-world context of patentability. In practice, this enables the trained architecture to classify new text pairs into X, Y, or A categories based on their semantic similarity, thereby emulating the decision-making process of patent examiners.

Design and training of the deep learning architecture

A deep learning architecture has been developed to classify patentability text pairs into X, Y, or A categories based on their semantic similarity, which is shown Fig. 1. The model will be trained on a dataset of tens of thousands of rows, each containing two text pairs and their corresponding patentability classification. Two approaches will be tested: (1) a CNN-based model using vectorized text pairs (e.g., via BERT) and (2) an RNN-based model

processing raw text directly. These alternatives will be compared to determine which offers the best accuracy and adaptability for patentability detection.

CNN architecture

In the CNN-based architecture, input consists of two 1×768 vectors from BERT, representing a pair of patent texts: one related to the current patent document under investigation and the other corresponding to the cited document. These vectors are either stacked into a 2×768 matrix for processing by a 2D CNN or directly compared. The CNN applies convolutional layers, followed by max pooling, batch normalization for stabilizing, flattening, and a fully connected layer. A softmax activation then classifies text pairs into three categories: X (high similarity, lacks novelty), Y (moderate similarity, inventive step concerns), or A (low similarity, patentable). The model is trained on thousands of labeled pairs to predict the X, Y, or A category for new patent text pairs. The overall design of the CNN model is illustrated in Fig. 3.

Siamese CNN architecture

In the second approach, a Siamese network with twin 1D CNNs processes the two 1×768 BERT-generated vectors separately. Sharing identical weights and structure, the twin CNNs independently extract features from each input. Each branch employs convolutional and pooling operations similar to those in the CNN model described in Sect. 3.4.1, but applied independently to each input vector. The extracted feature representations are then compared by computing the absolute difference between them using a Lambda layer, which serves as the similarity measure. The overall design of the Siamese CNN model is illustrated in Fig. 4.

This symmetric design ensures that the model captures nuanced relationships between patent texts, making it particularly effective for comparing claims and descriptions without combining the vectors into a matrix, as in the first method. The key methodological distinction lies in the parallel processing branches and the explicit distance calculation at the final stage. During testing, the Siamese CNN demonstrated superior performance over the 2D CNN in determining patentability, highlighting its ability to independently model and compare separate patent vectors more effectively. Based on these results, further optimization of the Siamese architecture will involve fine-tuning hyperparameters to enhance performance.

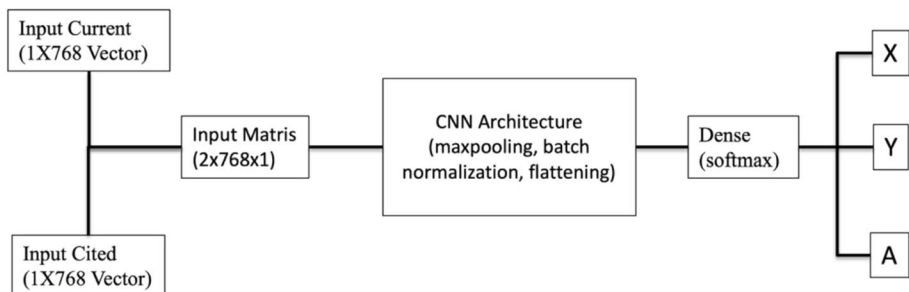


Fig. 3 Overall design of the CNN-based architecture

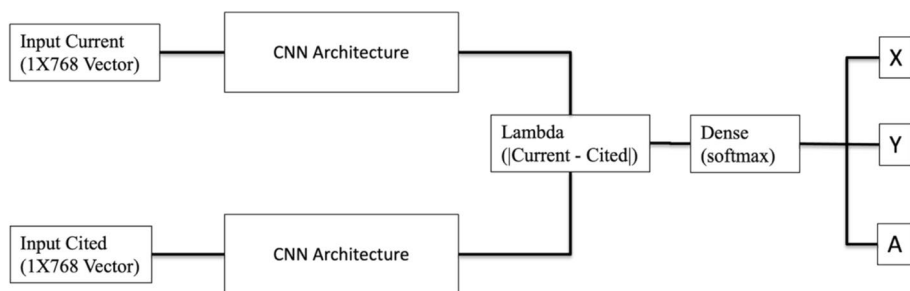


Fig. 4 Overall design of the Siamese CNN-based architecture

Siamese LSTM architecture

The third approach leverages an RNN-based method, specifically a Siamese LSTM architecture, to process patent texts directly without relying solely on fixed-size embeddings. Each input text is represented through an embedding layer and then passed to LSTM units that capture sequential dependencies across words. To mitigate overfitting, dropout regularization is applied after the LSTM layer, followed by dense projections for feature extraction. The outputs of the twin networks, which share weights, are compared using a distance function implemented via a Lambda layer, and the resulting vector is classified into the X, Y, or A categories through a softmax activation layer. This methodological design enables the model to capture temporal and contextual patterns in the text, providing an alternative to CNN-based approaches. The general workflow of this Siamese LSTM model is shown in Fig. 5, while its detailed instantiation and empirical results are presented in Sect. “[Siamese LSTM architecture](#)”.

Experiments, results and discussions

In this section, efforts were first made to generate a patentability dataset for testing the designed model. A unique and highly promising patentability dataset in the field of electric cars was obtained and shared with researchers. Then, patentability text pairs were derived from the patent texts in this dataset. Various deep learning architectures were trained with these text pairs, and the results were compared. Additionally, several factors that may affect

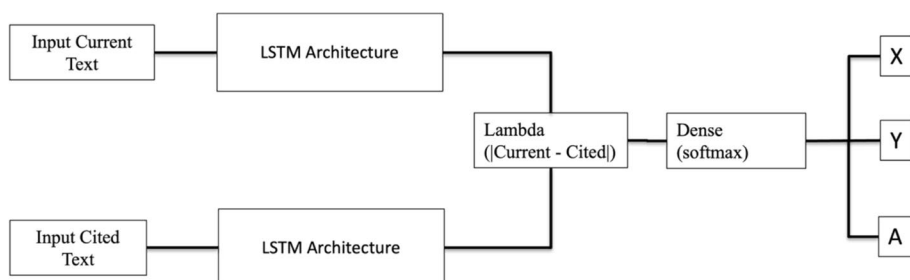


Fig. 5 Overall design of the Siamese LSTM-based architecture

Formation of test dataset in the field of electric car

In this test study, the professional patent software EPOQNET was used to generate the dataset on electric cars. A document set was obtained using the query “electric + and (vehicle + or car +)” in the title field of this software. This query was limited to applications published between 2000 and 2023, meaning it is related to technologies from the recent past. However, not every published patent has search report information available. To access patents with search reports, European patents with the publication number EP and international patents with the publication number WO were filtered. Since search report information is not available in every European or international patent publication, publications with codes A1 and A3, which contain this information, were also filtered. To perform accurate text analyses within the obtained set, applications with English-language description texts were selected.

As a result, a patent set containing 7941 patent documents having search reports was obtained. Using the XML outputs of the EPOQNET program, the description and claim texts of the retrieved patent documents, as well as the description texts of the cited patents in the search reports, the relevant claim numbers, and similarity categories were obtained. A snapshot of the resulting dataset can be seen in Fig. 6. The patentability dataset on electric cars generated within the scope of this study is shared in JSON format in the link below, with the aim of being useful to researchers interested in this topic (Kayakökü & Tüfekci, 2024).

Current Patent	Title	Abstract	Description	Claim Num	Claim Text	Similarity Category	Cited Patent	Cited Title	Cited Abstract	Cited Description	Cited Claims
52	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	1	to the motor winding	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
53	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	2	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
54	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	3	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
55	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	4	of the three-phase br	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
56	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	5	capacitors connected i	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
57	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	6	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
58	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	7	ge arm of the three-ph	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
59	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	8	is connected to the br	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
60	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	9	alternating current o	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
61	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	10	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
62	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	11	ms to 1.2 s, wherein th	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
63	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	12	version apparatus acc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
64	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	13	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
65	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	14	me of claims 1 to 11, th	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
66	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	15	to the motor winding	A	WO020224994	INTEGRATED CHARGING According to one aspe	CIATED CHARGER (The apparatus of claim 1		
67	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	16	in connection and disc	A	WO020226748	ELECTRIC VEHICLE AND Disclosed are an el	ectric vehicle and its (An integrated controller		
68	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	17	ected to the motor win	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
69	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	18	of the three-phase br	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
70	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	19	capacitors connected i	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
71	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	20	in connection and disc	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
72	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	21	ge arm of the three-ph	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
73	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	22	is connected to the br	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
74	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	23	in connection and disc	A	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
75	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	24	to the motor winding	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
76	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	25	g alternating current o	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
77	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	26	in connection and disc	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
78	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	27	ms to 1.2 s, wherein th	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
79	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	28	version apparatus acc	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
80	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	29	in connection and disc	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
81	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	30	me of claims 1 to 12, wh	X	US029625905	CHARGING DEVICE For a charging device for a	FEEL (0001) The inve (A charging device for a m		
82	EP0470450	ENERGY CONVERSION This application pro	TECHNICAL FIELD (0001)	31	in connection and disc	X					
83	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	1	ive, post, and a con	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
84	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	2	r fine iron is heedy con	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
85	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	3	in connection and disc	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
86	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	4	end, facing away from	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
87	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	5	er circumferential pos	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
88	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	6	in connection and disc	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		
89	EP0468580	(A2) AJMOTOR, POWER (A2) A3) This applica	TECHNICAL FIELD (0001)	7	the two planes are di	A	US0201217585	(A1) B2)Grounding me; (A1) B2)An electric mo	TECHNICAL FIELD (0001) "What is claimed is," An		

Fig. 6 A snapshot of the patentability dataset on electric cars

<https://www.kaggle.com/datasets/6cfe66ce0f1548de3e7c32c832a5701f49ae5498ed38ee027391a8a5fdd8b36>

As can be seen above, the first column in the dataset contains the publication number of the patent under investigation for patentability, in other words, the patent for which a search report was prepared. The second column contains the title, the third column the abstract, and the fourth column the description text of the patent. The fifth column contains the number of the claim being focused on for patentability, and the sixth column contains the text of that claim. The seventh column includes the expert's evaluation regarding the patentability of this claim as X, Y, or A category. From the eighth column onwards, the dataset contains information related to the document cited from the prior art for the patentability comparison, including its publication number, title, abstract, description, and claims section.

As can be observed, in many rows where the same values are present in several columns, there is no row where the text of a specific claim, the similarity category of the claim, and the citation information from the prior art are all identical. Therefore, the distinctive feature of this dataset is that it has been generated with a focus on patentability evaluation.

Vectorization and comparison of patent texts

After constructing the patentability dataset, a series of preliminary experiments were conducted on a limited subset of 15 patents to evaluate alternative approaches for representing and comparing patent texts. These initial tests employed the Doc2Vec method combined with cosine similarity to determine whether semantically related patents could be identified and distinguished from unrelated ones. The results indicated that while Doc2Vec captured general thematic relationships—such as differentiating patents from unrelated technical fields—it was unable to reflect the expert-defined X, Y, and A classifications used in patentability assessments.

Following this, the BERT model was tested for its ability to generate more context-aware embeddings of patent text. BERT demonstrated superior performance in capturing the semantic alignment between a claim and its corresponding description, as well as between claims and cited documents. However, even BERT, when applied solely with similarity metrics like cosine distance, proved inadequate for reliably predicting patentability categories. These findings reinforced that unsupervised similarity-based methods alone are insufficient to model the complex legal and technical reasoning involved in patentability evaluations.

Based on these insights, BERT was selected as the core text embedding technique for the study—not as a standalone classifier, but as the input layer for a supervised deep learning architecture trained on real-world, expert-labeled patentability outcomes. The full methodology and results of these early embedding comparisons are presented in Appendix A, while the main model development builds upon the strengths of BERT within a neural classification framework.

Generation of patentability text pairs

After the patentability dataset was generated and vectorization method was decided, the next step was to extract the pairs of text from the long patent documents in the dataset that would be used for patentability comparison. As explained in the previous section, the claim

text was used as the reference for determining the segment of the description to be selected. This is because patentability evaluation is claim-focused. By selecting the paragraphs in the description that are closest in meaning to the claim, we can identify the segments of the description most relevant to patentability.

To achieve this, the vector for each claim in every row of the dataset was determined using the BERT-Base method. This vector was then semantically compared to each paragraph in the description individually. Each paragraph of the description was vectorized using the BERT method, and the cosine similarity between the claim's BERT vector and the BERT vector of each paragraph was calculated. All paragraphs in the description were then ranked from highest to lowest based on their similarity to the claim.

At this point, an important decision in the model design arises: Which paragraphs, based on their proximity to the claim, should be selected to have the most significant impact on the model's performance? Deciding what percentage of the description should be selected based on its similarity to the claim is a critical aspect of the model design. Is it more logical to take 10% of the description, 5%, or 20%? To answer these questions, a comparison was made using a test sample generated from the search reports of 100 patents in the dataset. The similarity between the claim and the description was analyzed by selecting various percentages of the description, starting from 1%. Ultimately, 100% was also included, meaning the entire description was considered as a whole.

The similarity scores between the selected segments of the description and the claim were compared using BERT+Cosine Similarity. The difference between the average cosine similarity value for documents with X similarity and the average cosine similarity value for documents with A similarity for each percentage of the description is presented in the table (Table 1):

In this table, the higher the X similarity ratio compared to the A similarity ratio, the better the performance of the process. When selecting segments of the description smaller than 10%, the X similarity average was significantly higher than the A similarity average. As the size of the selected description segments increased, this ratio decreased, and especially after 50%, the results became identical. This indicates that as the size of the description segments increases, it becomes more difficult to capture semantic differences.

Based on this table, while forming text pairs, the description segment size was chosen as 10%. In other words, the top 10% of the description most similar to the claim was selected, while the remaining 90% was excluded from the process. At this stage, the

Table 1 The relationship between the description segment ratio and the X/A ratio

Description Segment Ratio	Average Cosine Value of X Similarity	Average Cosine Value of A Similarity	X/A Ratio
% 3	0,897399989	0,8732	1,027714154
% 7	0,897935984	0,874049	1,027329299
% 5	0,897839566	0,874092	1,027168129
% 10	0,895067088	0,871536	1,026999366
% 15	0,89776039	0,874246	1,026896961
% 25	0,897827931	0,874383	1,026812746
% 50	0,897740591	0,874354	1,026747812
% 75	0,897740591	0,874354	1,026747812
% 100	0,897740591	0,874354	1,026747812

sections representing 10% of the description were taken based on the X/A criterion, and if further detailed analysis is needed, it is possible to select smaller sections from within this 10% portion.

To achieve this, Python code was written to extract segments from both the investigated description and the cited description, consisting of the paragraphs most similar to the first claim. In this test phase of the study, the process was carried out based on the first claim of the applications. This is because the first claim is the independent claim and contains all the essential elements of the application. Subsequent claims generally describe additional features related to the invention explained in the first claim. The patentability evaluation is initially carried out based on the first claim. Once a decision is made regarding the first claim, the dependent claims are evaluated. If the first claim's patentability evaluation results in an A classification, the dependent claims, which add new features, are also considered to be A. For these reasons, although the patentability dataset shared with researchers includes all the claims of the applications, only the first claims will be considered during this test phase.

With the operations performed on the newly generated patentability dataset, a new dataset was obtained. This dataset includes each application's first claim, the 10% segment of the description most similar to that claim, the 10% segment of the cited prior art description, the BERT vector of the first claim, the BERT vector of the description segment, the cosine similarity value between the claim and the description segment, the BERT vector of the cited description segment, and the cosine similarity value between the description segment and the cited description segment, as well as the cosine similarity value between the claim and the cited description segment. As a result, a new dataset of 24,000 rows, more focused on patentability analysis, has been obtained and a small portion of it shared below in Fig. 7.

Generation of training data

At this point, before moving on to the generation of the training data, the model must focus on cleaning outlier data. Otherwise, text pairs that could distort the data would be included in the training set. The model should use cosine similarity values to detect outlier data. During the processing of XML files exported from the EPOQNET program, it was observed that in some rows, instead of the first independent claim, a dependent claim linked to the first claim was included in the dataset. Due to the structure of dependent claims, it

Investigated Patent Number	Investigated Description Segment Text	Investigated Description Segment Vector	Investigated First Claim Text	Investigated First Claim Vector	Own Description Segment - Claim Cos. Similarity	Patentability Similarity Category	Cited Patent Number	Cited Description Segment Text	Cited Description Segment Vector	Investigated Description Segment - Cited Description Segment Cos. Similarity	Investigated Claim - Cited Description Segment Cos. Similarity
EP4088955	at a first contro	[-5.034]	A controllable	tensor([-4.750]	0.970830362	X	US2019202343	[0013] In so	tensor([-3.840]	0.944762647	0.936492443
EP4088955	at a first contro	[-5.034]	A controllable	tensor([-4.750]	0.970830262	X	US2020233411	[0104] The tri	tensor([-3.189]	0.937465727	0.924586058
WO201603704	[0012] In an ex	tensor([-3.064]	A full duplex	da tensor([-2.880]	0.943347692	A	EP0681199	The present inv	tensor([-3.944]	0.949621379	0.897115827
WO201603704	[0012] In an ex	tensor([-3.064]	A full duplex	da tensor([-2.880]	0.943347692	Y	US201124875	[0037] Accor	tensor([-2.917]	0.933815897	0.897432029
WO201603704	[0012] In an ex	tensor([-3.064]	A full duplex	da tensor([-2.880]	0.943347692	Y	US201128534	[0189] FIG. 2:	tensor([-2.793]	0.941448867	0.894474804
WO201603704	[0012] In an ex	tensor([-3.064]	A full duplex	da tensor([-2.880]	0.943347692	Y	US201301570	[0093] In em	tensor([-2.008]	0.91916126	0.849932909
WO201603704	[0012] In an ex	tensor([-3.064]	A full duplex	da tensor([-2.880]	0.943347692	Y	US201315455	[0102] FIG. 6:	tensor([-3.170]	0.925625086	0.874605596
EP4083540	[0019] As sho	tensor([-4.717]	An electric vehi	tensor([-2.727]	0.954511464	X	KR200400694	That is, in the	tensor([-5.621]	0.945663386	0.916557269
EP4083540	[0019] As sho	tensor([-4.717]	An electric vehi	tensor([-2.727]	0.954511464	X	US200909500	[0004] DE 10	tensor([-4.649]	0.948581696	0.914301574
EP4083540	[0019] As sho	tensor([-4.717]	An electric vehi	tensor([-2.727]	0.954511464	X	DE102018201	[0037] Im AC;	tensor([-8.066]	0.718784094	0.654260039
EP4083540	[0019] As sho	tensor([-4.717]	An electric vehi	tensor([-2.727]	0.954511464	X	DE102019203	;- enim AC; un	tensor([-8.259]	0.717884958	0.650804698
EP4079132	a data process	tensor([-4.758]	Control unit	(3C tensor([-4.302]	0.976482511	A	US201600000	[0028] FIG. 2:	tensor([-3.156]	0.924398243	0.912351251
EP4079132	a data process	tensor([-4.758]	Control unit	(3C tensor([-4.302]	0.976482511	X	US202003338	[0005] The fo	tensor([-3.107]	0.947635174	0.92788434
EP4079132	a data process	tensor([-4.758]	Control unit	(3C tensor([-4.302]	0.976482511	A	US202007340	[0120] Still re	tensor([-2.155]	0.953763485	0.929686665
EP4074587	[0008] The mi	tensor([-6.310]	An electric vehi	tensor([-6.366]	0.959405422	Y	US201522939	[0022] FIG. 3:	tensor([-6.239]	0.945859432	0.935338318
EP4074587	[0008] The mi	tensor([-6.310]	An electric vehi	tensor([-6.366]	0.959405422	Y	CN106672140	[0048] A seco	tensor([-6.335]	0.916213572	0.919272244
EP4074587	[0008] The mi	tensor([-6.310]	An electric vehi	tensor([-6.366]	0.959405422	A	US201731338	[0048] A seco	tensor([-6.335]	0.916213572	0.919272244
EP4074587	[0008] The mi	tensor([-6.310]	An electric vehi	tensor([-6.366]	0.959405422	A	US201918823	[0034] System	tensor([-5.525]	0.927611291	0.917821646
EP4074587	[0008] The mi	tensor([-6.310]	An electric vehi	tensor([-6.366]	0.959405422	Y	US202011475	[0837] The sy	tensor([-5.745]	0.934212267	0.9383834

Fig. 7 The new dataset after text pairs generation

was not possible to use the same method described above for generating pairs of patentable texts. As a result, for rows containing dependent claims, the cosine similarity between the description sections generated from the corresponding claims and the referenced text was noticeably low. Therefore, these rows were removed from the dataset. Secondly, some of the cited documents were in languages such as German or French. Consequently, the similarity between the description sections generated from these documents and the target claim was significantly lower compared to other similarity values. These rows were also removed from the dataset. Additionally, some description cells were found to be empty. These rows were identified based on abnormally low similarity values and subsequently removed from the dataset.

In examining this dataset generated for electric cars, it is possible to clean the outlier data by removing rows where the similarity between the claim and its own description is below 90%, or the similarity between the claim and the cited description is below 80%. The fact that rows below these thresholds genuinely contain issues that could distort the data proves the accuracy of the vectorization and text pair comparison processes carried out so far. This ability to easily detect outlier data represents an important stage in the model design.

After performing the outlier data cleaning as described, approximately 17,845 rows of data remained in the dataset. However, there is one more preprocessing step that needs to be performed. In the patentability similarity category column, there are codes such as D, E, L, P, and T, which appear in patent search reports but do not pertain to patentability evaluations and instead represent other types of relationships, such as the dates of the compared documents. Below is a table showing the number of rows in this dataset, which focuses solely on first claims, where these codes appear (Table 2):

Rows with D, E, L, P, and T codes in this table were deleted from the dataset, as they do not contain any information related to patentability. The I code, on the other hand, pertains to documents reported as X in patent search reports. For this reason, all I categories were revised as X. Consequently, the number of rows remaining in the dataset, cleaned of outliers and containing qualified data, is as follows (Table 3):

No stratification or artificial balancing was applied to the class distribution during the training process. This decision was made deliberately to preserve the natural proportions and patterns of expert decisions, which we consider critical for developing a model that reflects the complexity of real-world patentability assessments. Although this introduces

Table 2 Frequency of dataset category information

Patentability Assessment	Number of Occurrence
A	8139
D	1
E	64
I	1052
L	2
P	16
T	5
X	4695
Y	3871
Total	17,845

Table 3 Frequency of dataset category information after preprocessing

Patentability Assessment	Number of Occurrence
A	8139
X	5757
Y	3871
Total	17,757

some class imbalance, it ensures the training data remains representative of actual patent examination practices.

In constructing the training dataset, the concept of patentability text pairs was introduced to emulate how expert patent examiners evaluate novelty and inventive step. Rather than analyzing full documents, examiners typically focus on the first independent claim and compare it with the most relevant sections of the cited patent’s description. Based on this approach, two types of semantically meaningful text pairs were constructed: (1) the first independent claim of the investigated patent paired with the most relevant cited paragraphs, and (2) a reconstructed segment from the investigated patent’s own description—designed to reflect the scope of the first claim in another way—paired with the relevant cited paragraphs. The methodology for identifying these relevant segments is described in Sect. “[Generation of patentability text pairs](#)”.

Therefore the cleaned dataset of 17,757 rows served as the basis for generating two types of patentability text pairs used to train the deep learning models. Patentability can be decided by comparing the investigated claim with the cited description segment, or by comparing the relevant segment of the investigated description with the cited description segment. Both types of text pairs have been generated in a way that is conducive to learning patentability decisions. As a result, from the cleaned dataset, we have obtained two different text pairs and their corresponding patentability decisions, generating a 35,000-row training dataset that can be used to train the deep learning architecture. The text pairs generated as described above, along with their BERT vectors and the category information indicating the similarity, were obtained as training and test data as follows:

The inclusion of both types of text pairs in the training dataset serves to support the model in learning how patent examiners conduct nuanced comparisons. Each pairing type offers a complementary view of the claim scope and its relation to prior art. Their combined use is intended to enrich the model’s training by providing semantically diverse yet contextually aligned examples. More importantly, this curated dataset is used consistently across all deep learning architectures evaluated in this study. This ensures that performance differences can be attributed to model design rather than variations in training data. The primary aim of this dual pairing strategy is not to claim definitive performance gains, but to establish a standardized basis for fair and meaningful comparisons between alternative deep learning approaches, with the goal of proposing an effective model for patentability assessment.

Although the input features for the model are based on patent text segments, the classification output is not derived from superficial similarity scores. Instead, the model is trained using expert-labeled patent search reports, in which domain professionals assign X, Y, or A categories after considering novelty and inventive step criteria. These labels inherently reflect the technical and legal reasoning of patent examiners. As such, the model learns to approximate expert-level assessments based on real-world decision patterns, rather than relying solely on textual closeness.

Training and results of the deep learning architectures

Using the patentability evaluations derived from expert opinions in patent search reports on electric cars and the related text pairs, deep learning architectures were trained. At this stage, different deep learning architectures were tested to measure their performance in making patentability decisions. As seen in the Fig. 8 above, there are two types of input data available for training with text pairs. First, the vectorized representations of the texts via BERT, and second, the texts themselves can serve as training data. CNN-based architectures are suitable for training with vectorized data, while RNN-based architectures are appropriate for training with the raw text data. At this stage, work was first done on CNN-based models, and then the LSTM model, an RNN-based method, was tested. The architectures were trained using 80% of the 35,000-row training dataset, and the trained models were tested with the remaining 20% of the data. This 80/20 train–test split was consistently applied across all experiments to ensure comparability of results.

CNN architecture

For the CNN-based experiments, we implemented the architecture described in Sect. 3.4.1 with specific hyperparameter configurations tailored for training and evaluation. The convolutional layer was configured with 128 filters and a 3×3 kernel, followed by MaxPooling and BatchNormalization. The resulting features were flattened and passed through a Dense layer with 128 neurons before the softmax layer for 3-class classification (X, Y, or A). The experimental configuration of the CNN model is illustrated in Fig. 9.

The model was trained over 30 epochs using the Adam optimizer and the Categorical Cross-Entropy loss function. Upon completion of training, the classification performance was evaluated on the test set, and the corresponding accuracy metrics, classification report, and training–validation performance curves were obtained as presented (Table 4).

With the training of the model, an accuracy of over 50% was achieved in predicting the three-class classification problem. While this accuracy is a value that could assist in patentability predictions, further tests were conducted on other deep learning architectures to achieve higher accuracy rates. Furthermore, as seen in Fig. 10, the validation loss has not

Desc/Claim Text	Desc/Claim Bert	Cited Text	Cited Bert	Similarity
A road for charging electric ve	[-2.70237774e-]	For the current supply system of t	[-2.61503160e-01	A
electrical power, and the elect	[-5.32302856e-]	This then should enable the ene	[-3.49705875e-01	A
comprising a thermal energy e	[-3.52737367e-]	On the outlet sides of the conden	[-3.75761628e-01	Y
In act, the electrical power co	[-3.09729010e-]	Furthermore, automotive heatin	[-3.92410219e-01	X
In the first aspect, the inventi	[-4.56093878e-]	Said on the side of the hood po	[-3.73440802e-01	X
SUMMARY The present applic	[-4.25665885e-]	Such results are obtained with the	[-6.50288641e-01	X
A circuit assembly can includ	[-2.62608409e-]	FIG. is a section of the circuit moc	[-8.50596745e-03	Y
In order to achieve the obje	[-4.52820122e-]	A socket made of any suitable ins	[-3.47697318e-01	A
Embodiments of the present	[-6.39233947e-]	Approaches in accordance with	[-3.59702468e-01	A
A system is disclosed includ	[-4.34664369e-]	An improvement is described for c	[-3.73185456e-01	Y
In order to achieve the obje	[-6.80021107e-]	Bristles are disposed toward the c	[-5.62040925e-01	A
A system according to the i	[-6.11845136e-]	Additionally the electrolyte is very	[-3.90200943e-01	X

Fig. 8 A snapshot of test data for deep learning architectures

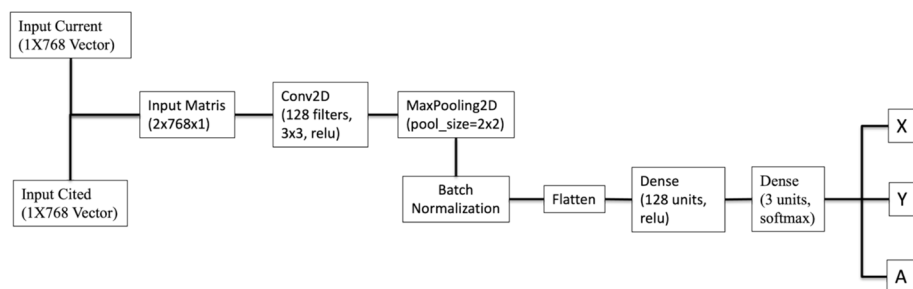


Fig. 9 The CNN architecture used in the experiment

Table 4 The test performance of the CNN architecture

Metric	A	X	Y	Macro Avg.	Weighted Avg.
Precision	0.65	0.56	0.45	0.55	0.58
Recall	0.65	0.54	0.49	0.56	0.58
F1-Score	0.65	0.55	0.47	0.56	0.58
Support	3282	2329	1479	7090	7090
Overall Test Accuracy: 57.7%					

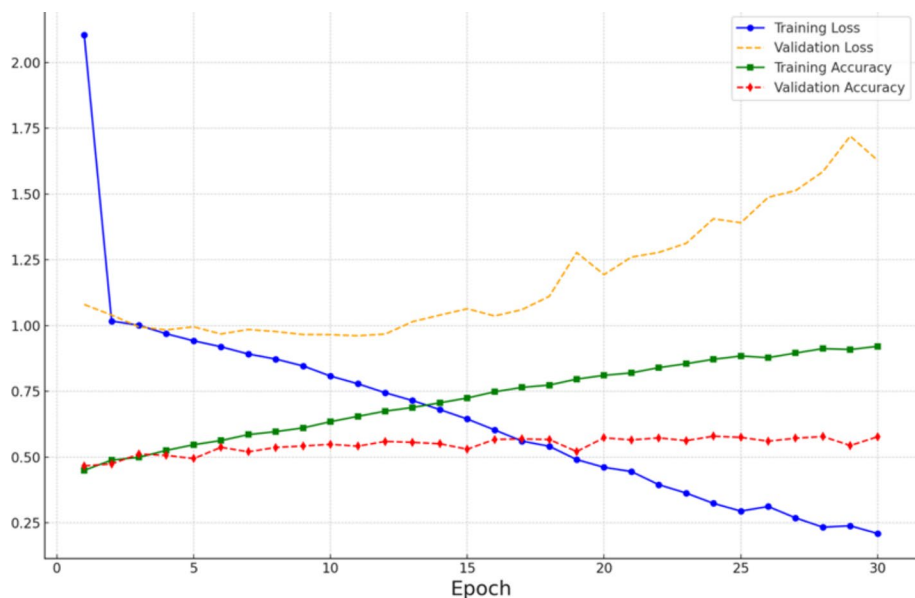


Fig. 10 The graph of training and validation metrics for CNN architecture

lost its upward trend, indicating that the losses are likely to continue increasing. Experiments were conducted further to develop a deep learning architecture that would exhibit a more stable trend in terms of both validation accuracy and validation loss.

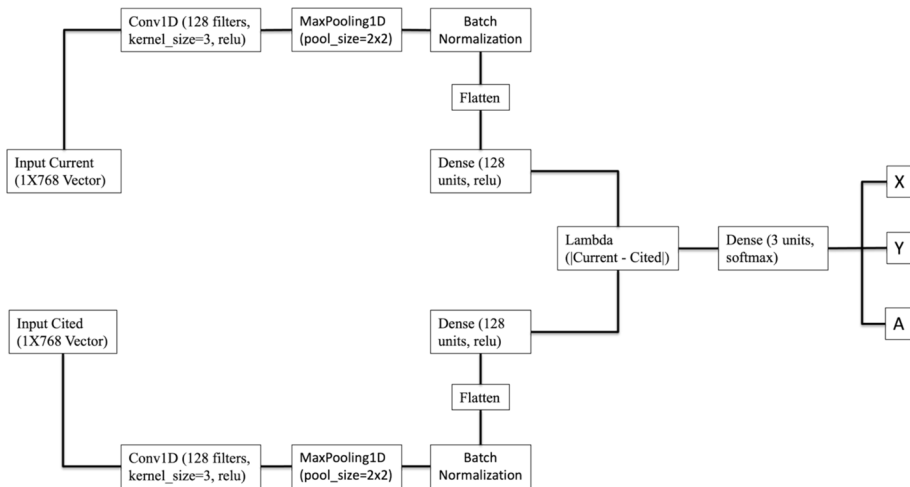


Fig. 11 The Siamese CNN architecture used in the experiment

Table 5 The test performance of the Siamese CNN architecture

Metric	A	X	Y	Macro Avg.	Weighted Avg.
Precision	0.67	0.64	0.55	0.62	0.63
Recall	0.74	0.58	0.49	0.60	0.63
F1-Score	0.70	0.61	0.52	0.61	0.63
Support	3282	2329	1479	7090	7090
Overall Test Accuracy: 63.6%					

Siamese CNN architecture

As introduced in Sect. 3.4.2, the Siamese CNN consists of twin 1D-CNN branches with shared weights that process two 1×768 BERT embeddings, one from the examined current document, one from the cited document. In this experimental instantiation, each branch applied a 1D convolutional layer with 128 filters, followed by 1D max-pooling, batch normalization, and flattening, similar to the setup in the 2D CNN model described in 4.5.1. The resulting feature vectors were then projected into a Dense layer with 128 units. The distance module employed the absolute-difference operator, as outlined in Sect. 3.4.2, and the final classification was performed by a Dense layer with softmax activation into the three patentability classes (X, Y, A). The instantiated architecture is illustrated in the Fig. 11.

The model was trained over 30 epochs using the Adam optimizer and the Categorical Cross-Entropy loss function. Upon completion of training, the classification performance was evaluated on the test set, and the corresponding accuracy metrics, classification report, and training–validation performance curves were obtained as presented (Table 5).

As can be seen from the results, the use of the Siamese network led to a significant increase in accuracy for both the training and test data. Additionally, while the accuracy reached around 0.90 after 10 epochs using the Siamese network, it took 28 epochs to reach

this level with the 2D CNN network. The graph in Fig. 12 indicates that the validation accuracy in the Siamese architecture shows a more stable increase, while the validation loss gradually plateaus after a certain point. Furthermore, in the classification report, there was a notable improvement in the precision, recall, and f1-score for all classes when using the Siamese network. For these reasons, it became clear that using a Siamese network for patentability determination in our study would yield more effective results. From this point onward, new tests were initiated to improve the performance of the Siamese network.

To further increase the accuracy of the Siamese CNN model, which was found to show better performance in patentability detection, efforts were made to implement certain improvements in the model design. First, it was examined whether passing the input vectors through the Conv1D+MaxPooling1D+BatchNormalization layer that extracts features from the input vectors in each branch of the Siamese network multiple times (sequentially) would help the network learn semantic patterns more effectively. In the architecture design shown below, for each case where the processing layer on both sides of the network had between 1 and 7 layers, the accuracy and loss values on the test data were measured after 50 epochs. The results were as follows (Table 6):

As can be seen from the results in the table, the lowest loss value and the highest accuracy were achieved with the architecture consisting of 5 processing layers on both sides of the Siamese network. As the number of layers increased up to 5, accuracy also increased, but after 5 layers, accuracy began to decrease as the number of layers continued to increase. For this reason, it was decided to set the number of processing layers in the model to 5. Therefore, the Siamese CNN architecture to be used by the model will be as follows (Fig. 13):

Another test on the model architecture was conducted regarding the filter size of the one-dimensional convolutional network. The filter sizes of the convolutional layers in each processing layer were set to 32, 64, 128, 256, and 512, and the accuracy values

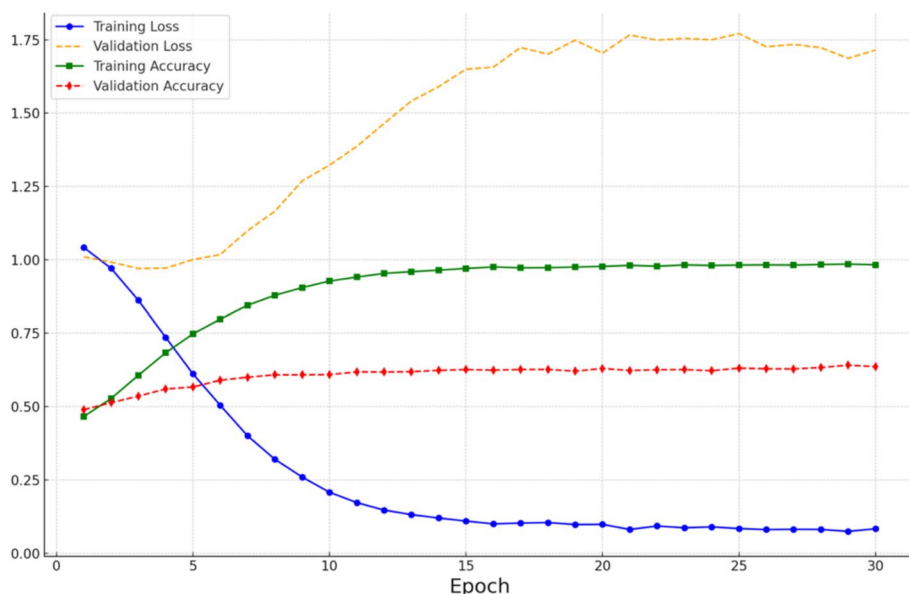


Fig. 12 The graph of training and validation metrics for Siamese CNN architecture

Table 6 The impact of the number of processing layers in the Siamese CNN architecture on accuracy and loss

Processing layer number	Test—Loss	Test—Accuracy
1 Layer	1,621,102	0,635,543
2 Layer	1,268,284	0,661,213
3 Layer	1,165,026	0,669,676
4 Layer	1,134,433	0,675,458
5 Layer	1,100,612	0,68,591
6 Layer	1,287,181	0,663,611
7 Layer	1,268,842	0,658,674

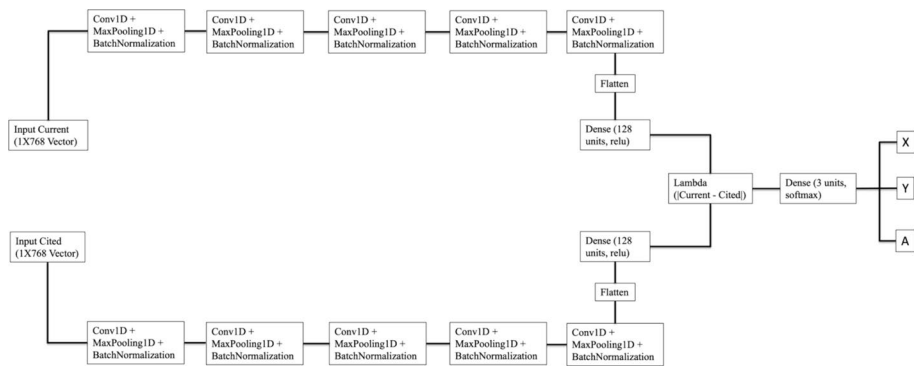


Fig. 13 The finalized Siamese CNN architecture

Table 7 The impact of convolutional filter size on accuracy and loss

Conv1D filter size	Test—Loss	Test—Accuracy
32	1,775,002	0,610,578
64	1,424,432	0,646,262
128	1,274,944	0,665,867
256	1,101,336	0,692,934
512	1,088,866	0,678,843

were checked over 50 epochs. The values obtained are shown in the table below. Since the highest accuracy was achieved with a filter size of 256, this filter size was chosen for the model design (Table 7).

After deciding on these hyperparameters for the model architecture, further tests were conducted on the generation of text vectors to try and increase accuracy. First, the impact of preprocessing steps on the accuracy of patent texts was investigated. When the test data above was generated, patent texts were vectorized in their original form, without any preprocessing. At this point, potential preprocessing alternatives that could be applied to patent texts and examined for their effect on the results are removing stop words and performing stemming to obtain the root forms of words, following classical text preprocessing steps; removing numbers to eliminate frequent reference numbers

Table 8 The impact of text preprocessing methods on accuracy and loss

Type of training dataset	Test—Loss	Test—Accuracy
Original patent texts	1,322,319	0,650,494
Patent texts with numbers removed	1,316,472	0,649,499
Patent texts retaining only keywords	1,7483	0,588,127
Patent texts processed with classical preprocessing steps	1,591,229	0,594,101

Table 9 The impact of embedding methods on accuracy and loss

Embedding method	Test—Loss	Test—Accuracy
BERT-base	1,238,896	0,665,162
BERT for patents	1,092495	0,680,154

found in patent texts; focusing on keywords by identifying the main keywords in the patent texts and concentrating on processing those.

These three alternative preprocessing steps were applied to the patent texts in the 35,000-row training dataset, and the resulting texts were vectorized again using the BERT method. The three alternative training datasets generated through these preprocessing methods, along with a fourth dataset that used the original patent texts without any preprocessing (which was used in the previous tests), were each used to train the Siamese CNN model for 30 epochs. The accuracy results on the test data for each dataset were as follows (Table 8):

As seen in the table, the highest accuracy was achieved when the patent texts were left in their original form. This shows how important it is to focus on processing patent texts in their original state, where every word, including even terms like “a,” “an,” and “the,” is carefully chosen and impacts the meaning. Applying classical preprocessing steps to patent texts, or focusing solely on keywords, can make it harder to capture the meaning of the text.

Another interesting observation in the results is that removing numbers, intended to eliminate reference numbers and paragraph numbers that frequently appear in patent texts without affecting meaning, reduced accuracy instead of improving it. This is because removing numbers not only eliminated reference numbers but also removed other numbers that were important for understanding the text. Therefore, if preprocessing is to be done on patent texts, it must be done with extreme precision to avoid affecting the meaning of the text. If such precision cannot be ensured, it is better to process patent texts in their original form.

After deciding to use the original patent texts for processing, the effect of the text embedding method on the deep learning model’s accuracy was examined. There are various derivatives of the BERT method, which has shown great success in vectorization, based on different pretrained language models. Among these, there is a derivative called “BERT for Patents” available at <https://huggingface.co/anferico/bert-for-patents>, which is specifically trained on patent texts. It is important to check how vectorization using different models impacts accuracy. For this purpose, the same text pairs were vectorized using both the BERT method and the “BERT for Patents” method, and the Siamese CNN model was trained. The accuracy and loss values on the test data are as follows (Table 9):

Table 10 The test performance of the tuned Siamese CNN architecture

Metric	A	X	Y	Macro Avg.	Weighted Avg.
Precision	0.,74	0.74	0.67	0.72	0.73
Recall	0.81	0.67	0.64	0.71	0.73
F1-Score	0.78	0.70	0.65	0.71	0.72
Support	3204	2217	1573	6994	6994
Overall Test Accuracy: 73.0%					

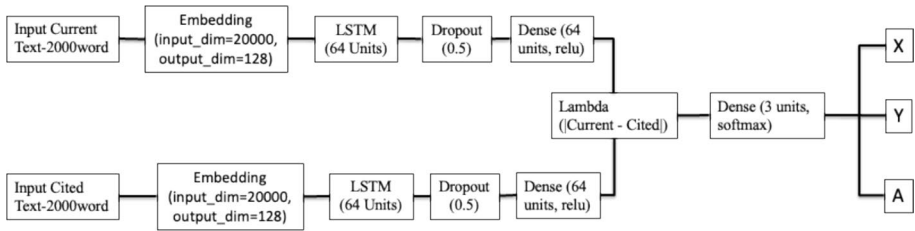


Fig. 14 Siamese LSTM architecture

As expected, when using the BERT method trained on patent texts (BERT for Patents), the model's accuracy improved. Therefore, after obtaining the patent text pairs, it was decided to vectorize them using the BERT for Patents method before training the Siamese CNN network.

By combining all these findings, the final approach involved vectorizing the original patent texts using the BERT for Patents method. The Siamese Network was designed with 5 layers, each containing a Conv1D network with a filter size of 256, and the model was trained for 150 epochs. The resulting accuracy and classification report are as presented (Table 10).

As can be seen from the results, the Siamese CNN network, which initially achieved an accuracy of 0.63, reached an accuracy of 0.73 after the improvements made to the same training data. These results demonstrate that the designed model has the potential to contribute to solving the patentability determination problem.

Siamese LSTM architecture

To evaluate the Siamese LSTM approach introduced in Sect. 3.4.3, a concrete architecture was instantiated and trained on the prepared dataset. Each input patent text was truncated or padded to a maximum length of 2000 tokens and mapped through an embedding layer limited to a vocabulary of 20,000 words. The embedded sequence was then passed through an LSTM layer with 64 units, followed by a dropout layer with a rate of 0.5 to reduce overfitting risks. The resulting features were further processed by a Dense layer with 64 neurons. The outputs of the twin networks were compared by computing their absolute difference via a Lambda layer, which was then classified into three categories (X, Y, A) using a Dense layer with softmax activation. The overall experimental architecture is depicted in Fig. 14.

Table 11 The test performance of the Siamese LSTM architecture

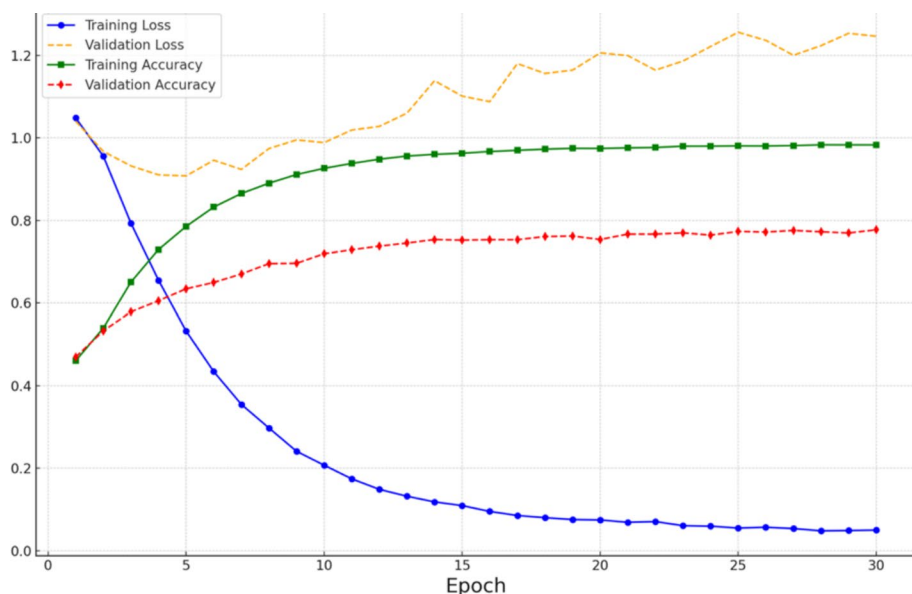
Metric	A	X	Y	Macro Avg.	Weighted Avg.
Precision	0.84	0.71	0.76	0.77	0.78
Recall	0.78	0.82	0.71	0.77	0.78
F1-Score	0.81	0.76	0.74	0.77	0.78
Support	3268	2245	1578	7091	7091
Overall Test Accuracy: 77.7%					

The model was trained for 30 epochs with “Adam” optimizer and “Categorical Cross Entropy” loss function, and the following accuracy values, classification report and the graph of training and validation metric value were obtained (Table 11) (Fig. 15):

As seen, the use of the Siamese LSTM network resulted in significant improvements in accuracy, precision, recall, and f1-score. According to the results, the Siamese LSTM architecture achieved higher accuracy compared to the Siamese CNN architecture. This demonstrates the potential of RNNs and their derivatives in patentability detection.

Comparison of deep learning architectures

In order to evaluate the effectiveness of different deep learning strategies in predicting patentability categories (X, Y, A), three distinct architectures were tested: a baseline CNN model using BERT-based vectorized inputs, a Siamese CNN model trained on semantically paired BERT embeddings, and a Siamese LSTM model that directly processes raw text sequences.

**Fig. 15** The graph of training and validation metrics for Siamese LSTM architecture

The performance of each model was compared using identical training and test datasets comprising 35,000 patentability text pairs. These pairs were generated using two strategies: (1) pairing the first independent claim with relevant cited paragraphs and (2) pairing reconstructed description segments (representing the claim's protection scope) with cited paragraphs. By keeping the dataset fixed across models, we ensured that performance differences stemmed from the model architectures and not the input data. It should also be noted that the inclusion of both types of text pairs within the same training dataset, while beneficial for exposing the model to diverse representations of patentability reasoning, may introduce a degree of semantic overlap that could potentially reduce the model's generalization capacity in broader applications. Therefore, the results should be interpreted primarily as relative performance comparisons between model architectures, with the goal of uncovering meaningful insights that contribute to the development of an effective patentability assessment model.

The baseline CNN architecture reached a test accuracy of 0.577, with a weighted F1-score of 0.58. While acceptable as a benchmark, the model exhibited signs of overfitting in later epochs, with training accuracy continuing to rise while validation accuracy stagnated. Furthermore, the recall value for the Y category remained relatively low, indicating difficulty in distinguishing semantically similar categories such as X and Y. This suggests that the architecture struggles to capture the nuanced distinctions required in patentability assessment.

In contrast, the Siamese CNN architecture achieved a test accuracy of 0.636 and a weighted F1-score of 0.63. The improvement reflects the model's better capacity to learn comparative relationships rather than relying solely on single input representations. Moreover, additional architectural tuning (e.g., increasing convolutional depth to 5 layers, using BERT for Patents embeddings) further enhanced performance to 0.73 accuracy. This shows that Siamese CNNs benefit significantly from well-prepared semantic embeddings.

The Siamese LSTM model, unlike the CNN variants, was trained using raw text as input. With an embedding layer trained on the corpus and a 64-unit LSTM on each side of the Siamese structure, this model reached a test accuracy of 0.777 and a weighted F1-score of 0.78, outperforming all other configurations. While the LSTM-based model had a much longer training time due to sequential processing and heavier computational cost, it demonstrated superior generalization, particularly in distinguishing subtle differences between categories X and Y. This suggests that LSTM-based models may better capture contextual nuance in unstructured patent text, especially when pretrained embeddings are not available or when raw syntax and sequential structure carry important meaning.

The key performance outcomes of all tested architectures are summarized in Table 12 to provide a comparative overview of their accuracy, efficiency, and input strategies.

In addition to final test metrics, the evolution of training and validation metrics over epochs provides further insights into the learning dynamics of each architecture. As shown in Fig. 10, the baseline CNN model demonstrates rapid overfitting: while training accuracy continues to rise steadily, validation accuracy plateaus early and validation loss increases after the mid-epochs. This divergence suggests limited generalization capability, especially for subtle patentability distinctions.

By contrast, Fig. 12 shows that the Siamese CNN architecture offers a more stable training trajectory. Validation accuracy improves steadily and validation loss exhibits a mild plateau rather than a sharp increase, indicating that the Siamese structure is better suited to capturing pairwise semantic relationships with reduced risk of overfitting. Notably, the performance gains become most pronounced after architectural tuning and the inclusion of domain-specific embeddings. Moreover, validation loss curves for the Siamese CNN

Table 12 Summary of comparative results

Model	Input type	Accuracy	Weighted F1	Training time	Notes
CNN	BERT vectors	0.577	0.58	Fast	Baseline, limited comparative ability
Siamese CNN (initial)	BERT vectors	0.636	0.63	Moderate	Better similarity modeling
Siamese CNN (tuned)	BERT for patents	0.73	0.72	Moderate	Best performance using pretrained vectors
Siamese LSTM	Raw text	0.777	0.78	High	Highest accuracy, but slower training

model are relatively smooth, reflecting a more stable optimization process with fewer fluctuations across epochs.

Figure 15, corresponding to the Siamese LSTM model, reveals a slower but more consistent improvement in both training and validation metrics. The model maintains a close alignment between training and validation accuracy throughout all epochs, with validation loss steadily decreasing. However, the validation loss trajectory shows more fluctuations compared to Siamese CNN, indicating a more sensitive training process. This can be attributed to the LSTM's sequential processing of raw text, which, while offering superior contextual understanding, also leads to higher susceptibility to noise or small variations in the training data. Despite this, the model exhibits strong generalization capacity and achieves the highest overall accuracy. However, this comes at the cost of significantly longer training times.

As a result, the experimental results indicate that Siamese LSTM provides the highest accuracy, likely due to its sequential processing of raw text, which captures contextual relationships better than static embeddings. However, the training process of the LSTM model exhibits higher sensitivity and less stability, as reflected in more fluctuating validation loss curves. This suggests that while the LSTM model excels in capturing semantic nuance, it may be more affected by variations in training data and requires careful tuning. In contrast, Siamese CNN architectures offer a good balance between performance and efficiency, especially when used with specialized embeddings like BERT for Patents. Their smoother optimization behavior and lower validation loss volatility make them appealing for scenarios requiring computational scalability and robustness. Therefore, when pretrained domain-specific embeddings are available and computational efficiency is a concern, Siamese CNN can be considered the preferred architecture. On the other hand, for maximum performance in understanding nuanced language within patent texts, Siamese LSTM remains particularly promising, provided that sufficient training resources are available.

Conclusions and future work

The model developed in this study and the findings obtained provide a significant contribution to addressing the patentability assessment problem. It has been demonstrated that designing a model that specifically addresses patentability evaluation—distinct from other semantic comparison tasks—by focusing on the criteria of novelty and inventive step is crucial for meeting the needs in this field. While the model's performance in achieving X, Y, and A categorization is not yet at the desired level, it represents an important starting point.

This study also highlights the critical value of published patent search reports as a resource for research in patentability assessment and natural language processing. These publicly accessible reports, free from ethical concerns, offer a robust dataset that can be utilized for various research purposes across different technical fields. As mentioned in the testing phase of this study, the patentability dataset developed for electric cars and shared in JSON format is a valuable resource that many researchers can benefit from. This dataset is not only a significant resource for semantic analysis but also provides important opportunities for researchers working in technology management.

A significant development in this study is the filtering of long patent texts to extract only the sections that directly impact patentability decisions, which leads to two highly efficient text inputs for the model. It would be extremely difficult to make a patentability comparison

using only the abstracts, claims, or descriptions of patent applications. Therefore, the approach of filtering text specifically for this decision has become a strength of the study. Furthermore, defining the claim under scrutiny as the filtering criterion aligns the design closely with real-world patent research, as patent examiners also evaluate the parts of the description that are most relevant to the claim when making patentability decisions.

The combination of the text pairs approach with the power of Siamese deep learning architecture and using these text pairs as input for Siamese networks, greatly enhanced the model's ability to make accurate decisions. With Siamese networks, it is possible to work with 1D inputs, reducing the computational resources required for handling multidimensional matrices and improving performance.

Although the findings presented in this study have made significant progress in solving the patentability problem, it is clear that there is still much work to be done in this area. To facilitate more efficient and faster progress in future studies, some recommendations should be made.

First, future studies should focus on improving the methods used to obtain patentability text pairs. While comparing the vectors of claims and paragraphs is a good starting point, efforts should be made to generate text pairs that better reflect the patentability evaluation with more accurate and concise texts. The more precise and specific the text pairs are, the higher the prediction accuracy will be.

Future studies should also explore the impact of different text pairing strategies for training dataset construction on model performance. Although the curated dataset constructed using two types of text pairs in this work enables meaningful comparisons between deep learning architectures, further experimentation is needed to assess how various approaches influence the model's absolute accuracy in patentability classification. Understanding these effects is essential to strengthen the generalizability and robustness of future models. While the current work presents an encouraging and concrete step toward tackling the challenging task of patentability assessment, it also highlights the complexity of the problem and the need for continued advancements in this field.

Besides, the impact of class imbalance on model performance warrants further investigation in future studies. To evaluate the robustness of patentability assessment models under more controlled conditions, stratified sampling strategies or synthetic balancing techniques may be employed. However, it is essential to preserve the integrity of expert-derived labels during such preprocessing to ensure consistency with real-world decision-making dynamics in patent examination.

In the future, more advanced models can be integrated into the decision-making module, making the design flexible and adaptable to new developments. Another flexible feature of the design is the ability to generate alternative text pairs. When generating patentability text pairs, the deep learning architecture evaluates the texts obtained from different sections of the description separately and presents any X or Y decision to the user. This feature helps to ensure that documents with X or Y concerns are not overlooked.

As noted in the testing phase, the tests were limited to the first claims in the dataset. Each application in the dataset contains many different claims, most of which are dependent on the first claim or subsequent claims. Future work could involve modeling these dependencies and incorporating the semantic analysis and generation of patentability text pairs for all claims, which would allow for even more patentability-related data to be produced.

Another important direction for future work involves integrating a prior art retrieval component into the system. Although the current study does not address the retrieval stage explicitly, this step is often more straightforward in real-world scenarios, where researchers

can generate candidate sets of related documents using basic keyword and classification-based searches through public or commercial patent databases. The real challenge lies in the accurate evaluation of these documents for patentability—specifically, determining novelty and inventive step—which is the primary focus of this study. The proposed model, trained on expert-annotated patent search reports, enables automatic classification of such documents into X, Y, or A categories once they have been retrieved. Therefore, the model offers an operational solution to the most expertise-intensive part of the patentability assessment process. In future implementations, retrieval mechanisms based on semantic similarity and dense vector search can be incorporated to complete the end-to-end automation of the patentability evaluation pipeline, provided that access to full-scale international patent databases is made available.

In conclusion, the findings from this study represent a significant contribution to solving the patentability detection problem, a critical challenge in the field. The results also show that much more work is needed to fully solve this problem. Since patentability evaluation is a unique process shaped by patent terminology and legal regulations, future studies in this field should be specifically designed to address patentability. This study has outlined the unique characteristics of patentability and demonstrated that progress can be made in solving this difficult problem with patentability-focused designs.

As deep learning, natural language processing, and semantic analysis methods continue to advance, their capabilities must be rapidly applied to patent documents and databases. It is clear that expertise in the patent system is also required to achieve this. Progress in this area will not be possible without combining patent knowledge with data science. Encouraging data scientists to work in the patent system and encouraging patent experts to explore data science will open the door to significant developments in the field of technology and innovation.

Appendix A: Preliminary comparison of embedding methods

Doc2Vec method

After obtaining the patentability dataset, the stages of text mining, vectorizing the documents, and similarity checks were initiated. To perform text mining operations and conduct preliminary design and testing of the model, a subset of 15 patents was selected from the dataset. Additionally, for each patent, only the first claim was considered, as the first claim of each patent typically contains all the essential elements of the invention in question.

The doc2vec method, a deep learning-based word representation technique, was used to vectorize lengthy patent texts. Using Python's Gensim library and a pre-trained doc2vec model based on English Wikipedia data available <https://github.com/jhlau/doc2vec> (DBOW approach, 1.4GB), 300-dimensional document vectors were generated to capture the semantic meaning of each patent. Cosine similarity between these vectors was then calculated to predict the X, Y, and A similarity categories assigned in search reports, focusing on descriptions and first claims of 15 patent applications and their cited patents.

In the Fig. 16 below, the data is ranked according to the similarity between the description of the investigated document and the description of the cited document firstly and then similarity between the first claim of the investigated document and the description of the cited document:

a

H

However, there are also highly meaningful results produced by the doc2vec similarity method. For instance, the values in column G of the tables above represent the similarity ratios between the investigated patent document's own description and its own claim. In other words, these claims are the ones already described within the document itself. The similarity ratios between these claims and their corresponding descriptions are significantly higher than the similarity ratios with the descriptions of the cited documents. For example, while the highest similarity ratio between the investigated claim and the cited description is 0.49, the lowest is 0.36, and the average is 0.42, the similarity ratios between the investigated claims and their own descriptions are as follows: the highest similarity ratio is 0.69, the lowest is 0.47, and the average is 0.54. This demonstrates that the doc2vec method shows that a claim is much more similar to its own description than to the descriptions of the cited documents from the search report.

At this point, it should also be noted that the relatively low similarity values are due to the use of a pre-trained doc2vec model from a different domain. If a doc2vec model trained on patent texts were used, these values would likely be higher. However, since we are conducting a relative analysis by comparing the values with each other, the absolute magnitude of the values does not affect the analysis results.

To further test the effectiveness of the doc2vec model in patent similarity analysis, the patent documents in the electric car field from the table above were compared with patents from unrelated fields. As previously explained, even if a document in the search report is categorized as A, it is still from the same field as the investigated patent and has similar technical features. Therefore, patents from different fields that would not even be included in the search report should have a noticeably lower similarity to the patents in the electric car field. To test this, patents from unrelated fields were compared with the patents listed in the figure below, as follows (Fig. 17):

As can be seen from the values in the Fig. 14, the similarity ratio between the patents in the field of electric cars and unrelated documents is significantly lower compared to the documents cited in the search report. While the similarity ratio with the cited patents is around 0.47 on average, the similarity ratio with unrelated documents is around 0.24 on average. The conclusion that emerges from this is that when patent documents are compared using the doc2vec model for semantic similarity, related documents from the same technical field can be identified. Although this aspect of the model alone is a development that could reduce the workload in patent research, the main goal here is to be able to establish the X, Y, and A categorization.

	A	G	H	I	J
1	Current Patent	Non Related Documets	Non Related Description	Des - Non Related Si	Claims - Non Related Si
2	EP4083540	US11412721B2	CROSS REFERENCE TO RELATED APPLICAT	0,110828534	0,175642893
3	EP4083540	US11412721B2	CROSS REFERENCE TO RELATED APPLICAT	0,135292917	0,120034792
4	EP4083540	US10220005B2	[0002] The present invention relates to t	0,155961975	0,231714144
5	EP4089816	US10718495B2	FIELD OF THE INVENTION[0001] The pres	0,173843622	0,217605993
6	EP4089816	US10596603B2	BACKGROUND OF THE INVENTION[0002]	0,176621005	0,155292422
7	EP4089816	US11412721B2	CROSS REFERENCE TO RELATED APPLICAT	0,18662037	0,217603609
8	EP3470280	US10596603B2	BACKGROUND OF THE INVENTION[0002]	0,189257517	0,237749159
9	EP3470280	US10220005B2	[0002] The present invention relates to t	0,190406531	0,205833063
10	EP3470280	US10575642B2	FIELD OF THE INVENTION[0002] Present i	0,199516729	0,13069585
11	EP3470280	US10718495B2	FIELD OF THE INVENTION[0001] The pres	0,202540919	0,191500589
12	EP4079132	US10718495B2	FIELD OF THE INVENTION[0001] The pres	0,203304335	0,20732227
13	EP4079132	US10220005B2	[0002] The present invention relates to t	0,207823634	0,241446301
14	EP4046858	US10718495B2	FIELD OF THE INVENTION[0001] The pres	0,212551519	0,172323078
15	EP4046858	US10220005B2	[0002] The present invention relates to t	0,213843912	0,236329913
16	EP4074540	US10596603B2	BACKGROUND OF THE INVENTION[0002]	0,214645922	0,150317967
17	EP4074540	US10596603B2	BACKGROUND OF THE INVENTION[0002]	0,224234492	0,227178738
18	EP4074540	US10406412B2	TECHNICAL FIELD[0002] The present inve	0,229231447	0,217123151
19	EP4074587	GB2568046A	SAUSAGE PRODUCING APPARATUSField c	0,236102313	0,272506356
20	EP4074587	US10220005B2	[0002] The present invention relates to t	0,2411834	0,268078268
21	EP4074587	US10220005B2	[0002] The present invention relates to t	0,253478795	0,23512432
22	EP4074587	US10575642B2	FIELD OF THE INVENTION[0002] Present i	0,253588557	0,184144497
23	EP4067704	US10406412B2	TECHNICAL FIELD[0002] The present inve	0,254374862	0,227124333
24	EP4067704	US10575642B2	FIELD OF THE INVENTION[0002] Present i	0,258035541	0,154168814
25	EP4067704	US10018353B2	TECHNICAL FIELD[0003] The present disc	0,259098987	0,265314668
26	EP4067198	US10718495B2	FIELD OF THE INVENTION[0001] The pres	0,259234548	0,259426415

Fig. 17 The similarity of the investigated patents with patents on irrelevant topics

BERT method

In recent studies in the literature, the Transformer-based BERT method has been highlighted for its significantly higher performance in text and content representation. Therefore, it was preferred to use the BERT method instead of doc2vec, with the expectation that vectorizing patent texts with BERT would yield more accurate results. Using Python's transformers library with BertTokenizer and BertModel, along with the torch library, all texts were vectorized with the pre-trained "bert-base-uncased" model. Since the BERT model can vectorize a maximum of 512 characters, long patent texts were split into segments of 512 characters. The vector for the entire document was then obtained by averaging the vectors of each segment.

To compare the doc2vec and BERT methods, the same similarity calculations were performed on the same sample, which was previously calculated using the doc2vec method, but this time using the BERT method. As seen in the results below in Fig. 18, the similarity ratio of patents vectorized using the BERT method was significantly higher than those vectorized using the doc2vec method.

Similar to the doc2vec method used above, the BERT method employed here is not a model specifically trained on patent texts but a general-purpose model. Therefore, it is possible to make a comparison between the two results. While the average similarity between a claim and its own description was 0.53 with the doc2vec method, this value increased to 0.91 with the BERT method. This is an excellent result, as each claim should semantically resemble its own description to a large extent. Similarly, the average similarity between a claim and the document it was compared with in the search report was 0.43 with the doc2vec method, whereas this value reached 0.88 with the BERT method. This was also an expected result since the documents cited in the search report are very closely related

Investigated Patent Number	Claim Number	Own Cos. Similarity	Cited Patent Number	Similarity Category	Desc.-Cited Cos. Similarity	Claim.-Cited Cos. Similarity
EP4079564	1	0.912763417	W02012075186	X	0.908077721	0.913810246
EP4089816	1	0.891099811	US2019081298	A	0.96764436	0.938834412
EP4089816	1	0.891099811	US2020388482	X	0.966821909	0.887156844
EP4079564	1	0.912763417	EP20589841	Y	0.962570369	0.912139926
EP4079564	1	0.89891088	CN109854688	A	0.91747473	0.938200219
EP4089816	1	0.891099811	US201833737	X	0.960791355	0.873997757
EP4079564	1	0.912763417	JP2013085388	X	0.960779732	0.930815756
EP4068808	1	0.958806872	US20257114	Y	0.906021417	0.923190766
EP4079564	1	0.912763417	US2014126606	A	0.960587025	0.920283496
EP4068808	1	0.891584111	US20191341661	Y	0.96023649	0.878748715
EP3470280	1	0.922816217	JP2002354603	X	0.939175858	0.937887774
EP4067198	1	0.906588137	US201810510	X	0.935369957	0.899151862
EP4067198	1	0.906588137	US2015120101	X	0.958618833	0.894128818
EP4068808	1	0.891584111	JP2017105290	Y	0.956071258	0.914528218
EP4079564	1	0.89891088	CN105113449	A	0.955605665	0.871502729
EP4079132	1	0.915066659	US202003386	X	0.954825499	0.936800575
EP4079564	1	0.912763417	US201228549	Y	0.953542556	0.854411653
EP4074540	1	0.924407005	W02020149944	A	0.952297449	0.924555179
EP4074540	1	0.924407005	US2019275905	X	0.95109117	0.890515313
EP4074587	1	0.936123431	US201250934	Y	0.950161609	0.89090129
EP4079564	1	0.89891088	US2020079190	A	0.947293435	0.913739622
EP4079132	1	0.915066659	US201600005	A	0.94820416	0.899176223
W02016037045	1	0.871514593	EP0681199	A	0.948153138	0.861856001
EP4074540	1	0.924407005	CN109187423	A	0.948141873	0.845661938
EP4074587	1	0.936123431	US201918235	A	0.947888384	0.87971158
EP4089816	1	0.891099811	EP4861001	X	0.945174846	0.850265321
EP4079564	1	0.912763417	US2016347638	Y	0.945127231	0.917153895
EP3470280	1	0.922816217	US20227368	A	0.944414377	0.910551667
EP4074587	1	0.936123431	US201731380	A	0.942393064	0.886570752
EP4074540	1	0.924407005	US201027648	A	0.94266047	0.800798837
EP407517	1	0.939884603	JP200808873	Y	0.940232853	0.931878075
EP3470280	1	0.922816217	US201707910	X	0.940543549	0.917574286
EP407517	1	0.939884603	US2016317149	Y	0.940191329	0.908164978
EP4068808	1	0.958806872	US2010127385	Y	0.939825058	0.880629897
W02016037045	1	0.871514593	US2011248754	Y	0.939738035	0.865454912
EP4068808	1	0.958806872	US2019131899	A	0.937846059	0.88711381
EP4067074	1	0.89891088	JP204621221	X	0.936020616	0.928552611
EP4089816	1	0.915066659	US2019202031	X	0.936069693	0.87118411
EP4079132	1	0.915066659	US202073403	X	0.934810877	0.860709588
EP4089816	1	0.915066659	W02016161256	X	0.934389234	0.876191139
EP4089816	1	0.900565803	US2013130516	X	0.930303514	0.829935157
EP4089816	1	0.915066659	US202023410	X	0.929206043	0.869970739

a

Investigated Patent Number	Claim Number	Own Cos. Similarity	Cited Patent Number	Similarity Category	Desc.-Cited Cos. Similarity	Claim.-Cited Cos. Similarity
EP3470280	1	0.922816217	JP2002154003	X	0.932856217	0.930174874
EP4079564	1	0.912763417	JP2013085388	X	0.960779732	0.930815756
EP407704	1	0.89891088	CN109854688	A	0.961764753	0.930200219
EP4089816	1	0.900565803	US2013130516	X	0.930303514	0.829935157
EP407704	1	0.89891088	JP2022166127	X	0.936020616	0.928459211
EP4074540	1	0.924407005	W02020149944	A	0.952297449	0.923451579
EP408580	1	0.958806872	CN121272144	Y	0.960621417	0.923390746
EP4079564	1	0.912763417	US2014126606	A	0.961763417	0.923390746
EP3470280	1	0.922816217	US2017079210	X	0.940191329	0.908164978
EP4079564	1	0.912763417	US201644286	Y	0.945127231	0.917513895
EP4079132	1	0.915066659	US202003386	X	0.954825499	0.936800575
EP4083540	1	0.84345805	US200909505	A	0.912148553	0.913739622
EP4068808	1	0.893584311	JP2017105290	Y	0.956071258	0.914528218
EP407704	1	0.89891088	US2020079190	A	0.949293435	0.913739622
EP4079564	1	0.912763417	W0202017586	X	0.960397721	0.913412046
EP4079564	1	0.912763417	EP2050941	Y	0.962570369	0.912339026
EP3470280	1	0.922816217	US20227368	A	0.944414377	0.910551667
EP4074587	1	0.936123431	US2015292934	Y	0.950361609	0.909900129
EP407517	1	0.939884603	US2016137149	Y	0.940191329	0.908164978
EP4068808	1	0.893584311	US2016173854	A	0.927562177	0.90680365
EP4079198	1	0.906588137	US2014180510	X	0.939369957	0.899151862
EP4089816	1	0.900565803	US2019176223	X	0.869339048	0.898961395
EP3470280	1	0.922816217	US2016295797	A	0.913414996	0.897882688
EP4079198	1	0.906588137	US2015120101	X	0.958618833	0.894128818
EP4089816	1	0.891099811	US2019081298	A	0.96764436	0.893844212
EP407517	1	0.939884603	JP2020088873	Y	0.940832853	0.89380765
EP4079564	1	0.912763417	US2016347638	Y	0.945127231	0.917153895
EP4074540	1	0.924407005	US2019275905	X	0.95109117	0.890515313
EP4089816	1	0.891099811	JP2015138449	X	0.966821909	0.887156844
EP408580	1	0.958806872	US2019131899	A	0.937846059	0.88711381
EP407517	1	0.939884603	US201731380	A	0.942393064	0.886570752
EP4083540	1	0.84345805	KR2004002695	X	0.913735058	0.82609094
EP408580	1	0.958806872	US201027648	Y	0.94266047	0.800629897
EP4074587	1	0.936123431	US2019138235	A	0.947888384	0.87971158
EP4068808	1	0.893584311	US2019141661	Y	0.96023649	0.878748715
EP407704	1	0.89891088	CN105113449	A	0.955605665	0.871502729
EP408955	1	0.913400337	W02016161216	A	0.934389234	0.876191139
EP4089816	1	0.915066659	US201833737	X	0.960791355	0.873997757
EP408955	1	0.913400337	US2020202831	X	0.936601963	0.87118411
EP408955	1	0.913400337	US202023410	X	0.929020643	0.869970739
EP4079132	1	0.915066659	US2020077340	X	0.934810877	0.860709588
W02016037045	1	0.873514593	US2011248754	Y	0.939738035	0.865454912

b

Fig. 18 Test set sorted based on BERT+cosine similarity. **a** According to similarity between the description of the investigated document and the description of the cited document. **b** According to similarity between the first claim of the investigated document and the description of the cited document

Table 13 Comparison of the doc2vec and BERT methods

	Similarity between claim and own description	Similarity between claim and cited description	Average cosine similarity for X category documents	Average cosine similarity for A category documents
doc2vec method	0.53	0.43	0.41	0.42
BERT method	0.91	0.88	0.89	0.88

in meaning to the claim. In this regard, the BERT method successfully captured these high similarities at a higher rate.

An important performance metric demonstrating the difference between the doc2vec and BERT methods is the ratio of the average similarity between the claim and the cited description in rows marked with X similarity in the dataset, compared to those marked with A similarity. The similarity between the claim and the cited document should be higher for documents categorized as X, as the research expert assigns the X code to the documents considered closest in terms of patentability. In this context, the average cosine similarity for claim-cited document pairs with X similarity in the doc2vec method was found to be 0.41, whereas the average similarity for pairs with A similarity was 0.42. In this case, the doc2vec method found A category documents to be more similar than X category documents, which is contrary to reality. However, when the same values were calculated using the BERT method, the average similarity for X category documents was 0.89, while the average similarity for A category documents was 0.88. This shows that the BERT method identified documents that are truly more similar to each other with a slightly higher similarity value, even in this small sample, compared to the doc2vec method.

The performance differences between the two methods are summarized in the table (Table 13):

This highlights the power of the BERT method and shows that processing semantically complex patent texts with this method is more logical. Indeed, while the doc2vec method generates 300-dimensional vectors, the BERT method generates 768-dimensional vectors. This means that the BERT method quantifies the meaning of the text in more detail. For this reason, the BERT method was chosen as the primary vectorization technique in the model design, and all vectorization operations were performed using this method.

However, despite the high similarity ratios obtained with the BERT method, when examining the results, it becomes apparent that the BERT method alone is still insufficient for deciding the X, Y, and A similarity categories. As shown above, when ranking the rows from highest to lowest based on cosine similarity, it is expected that the most patentable X-coded documents would cluster at the top, followed by Y-coded documents, and finally A-coded documents at the bottom. However, this distribution could not be achieved. Therefore, as stated in the initial model design, further developments to the model are needed to make patentability decisions.

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Declarations

Conflict of interests The authors declare that they are inventors on patent applications related to the methods described in this manuscript. No other financial or non-financial competing interests exist.

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